

# Conjugate heat transfer at an immersed interface: the scheme as implemented

As-built note — what the solver does today

## Abstract

This note describes the conjugate-heat-transfer scheme *as it exists in the code*. It is deliberately narrower than the design note `conjugate_ibm.tex`, which surveys three source methods, derives the exact cut-face flux including its tangential term, works out a corner model, and weighs alternatives. None of that survey is repeated here. What is described here is what runs: one face coefficient, a fluid-fraction-weighted capacity, a time-step bound, a configuration surface with five hard guards, and two diagnostics. Where a term exists in the code but is switched off by default, it is stated as such together with the measurements that decided it (§9); where a piece of the design was never built, it is listed so that the reader is not misled (§10).

## 1 What is solved

The solid stops being a boundary condition and becomes a real unknown. Fluid and solid carry the same scalar variable, advanced by the same kernel, and differ only through two ratios that are 1 in the fluid by definition:

$$\kappa = \frac{k}{k_f}, \quad C = \frac{(\rho c)}{(\rho c)_f}.$$

In the solver's non-dimensional form, with  $D_f = 1/(Re Pr)$  the fluid diffusivity, the transported scalar  $s$  obeys

$$C \partial_t s = -\nabla \cdot (\mathbf{u} s) + \nabla \cdot (\kappa D_f \nabla s) + C q, \quad (1)$$

with  $\mathbf{u} \equiv \mathbf{0}$  and  $\kappa = \kappa_s$ ,  $C = C_s$  inside the body. At the interface the physical conditions are continuity of temperature and of normal flux,

$$s_f = s_s, \quad k_f \partial_n s_f = k_s \partial_n s_s, \quad (2)$$

optionally relaxed to a contact resistance  $R_c$  (§3). **Neither condition is imposed anywhere in the code.** They are consequences of a single face coefficient, which is the whole scheme.

## 2 Geometry: one stored field

The scheme reads exactly one geometric field, and it is one the solver already produced for the turbulence models:

$$\varphi = \pm \text{dwall}, \quad (3)$$

the distance to the surface, *signed* negative inside the solid. It is ghost-inclusive and pointwise exact from the geometry — no halo exchange is needed for it, because every value is computed from the surface, not from neighbours.

The sign comes from the cell-centred immersed-boundary marker: a cell centre lies in the solid exactly when  $|\text{coef}(\text{VAR\_P})| > 10^{20}$ , because `set_ibm_coeff` writes `SOLID/Re` there and grades only fluid-centred cells. Both producers of `dwall` (the file-based path via the case file's `dwall_blocks`, and the analytic path) already supply what is needed; the conjugate mode adds **no new dataset and no case-file format change**.

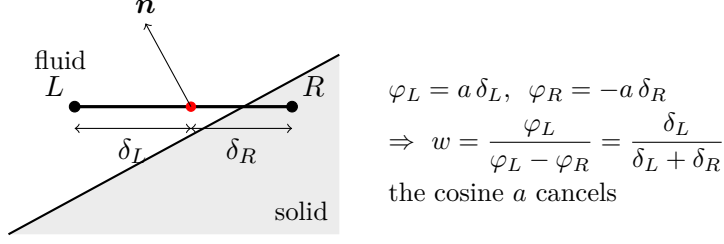


Figure 1: The obliquity lemma.  $\varphi$  stores the perpendicular distance, so the direction cosine  $a$  appears in both signed distances and cancels in their ratio:  $w$  is the arm fraction at any orientation, with no normal computed.

**The obliquity lemma.** This is why no normal vector is ever computed. Let the interface be locally a plane with unit normal  $\mathbf{n}$ , and consider a grid arm along  $\mathbf{e}_d$  joining two cell centres at distance  $h_d$ , with  $a = \mathbf{n} \cdot \mathbf{e}_d$  the direction cosine. Because  $\varphi$  stores the *perpendicular* distance,

$$\varphi_L = a \delta_L, \quad \varphi_R = -a \delta_R,$$

where  $\delta_L, \delta_R$  are the distances *along the arm* from each centre to the interface. The cosine therefore cancels in the ratio:

$$w = \frac{\varphi_L}{\varphi_L - \varphi_R} = \frac{\delta_L}{\delta_L + \delta_R} \quad (4)$$

is the true arm fraction *at any orientation*. That is the entire reason the scheme needs no new data.

### 3 The face coefficient — the whole scheme

At a face whose two cell centres straddle the interface ( $\varphi_L \varphi_R < 0$ ), the face diffusivity becomes the distance-weighted harmonic mean on  $w$ :

$$k_{\text{face}} = \frac{D_f}{\frac{w}{\kappa_L} + \frac{1-w}{\kappa_R} + R_c \frac{D_f}{h_d}} \quad (5)$$

Every other face keeps today's kernel line verbatim,  $k_{\text{face}} = \kappa_L D_f$ . That is the complete interface treatment: one sign test, one weight, one reciprocal, evaluated inside the flux line the kernel already had.

What it inherits:

- **Exact in one dimension.** It is the series resistance, so it is exact for a grid-aligned interface at *any* cut position, and exact at any orientation wherever the local gradient is interface-normal.
- **Conservative.** It is a face flux — the neighbour subtracts the same number — so  $\sum C s \Delta V$  is conserved to round-off.
- **Correct in both limits.**  $\kappa_R \rightarrow \infty$  recovers the isothermal wall,  $\kappa_R \rightarrow 0$  zero flux.
- **Inert when unused.** An uncut face takes the else-branch, which is the pre-existing arithmetic operand for operand, so a non-conjugate run is bit-identical.

**Grazing guard.**  $w$  is unreliable for arms nearly *tangent* to the surface, where  $\varphi_L - \varphi_R \rightarrow 0$ . The implementation therefore tests the direction cosine, which (4) supplies for free as  $a = |\varphi_L - \varphi_R|/h_d$ , and falls back to  $w = \frac{1}{2}$  below a threshold. Such arms carry little of the interface flux, so the fallback costs nothing measurable.

**Contact resistance.**  $R_c$  enters (5) as a third resistance in series, which is what it physically is. It is exact: a gate with  $R_c = 0.25$  reproduces the analytic two-material solution to 0.0.

## 4 The discrete scheme

### 4.1 Face loop

The six face diffusivities of a pressure cell are built with (5) and the flux divergence is formed as usual. Nothing else in the diffusion stencil changes.

### 4.2 Convection

Convection is hard-masked on solid *and* cut faces, and — this is the part that is easy to get wrong — the skew-symmetric term’s divergence is built from the *same* masked velocities. That is what keeps a uniform scalar exactly preserved, in the solid as well as the fluid.

### 4.3 Capacity at cut cells

The flux divergence is divided by the cell’s own volumetric capacity. A cut cell physically contains both materials, so the capacity is **fluid-fraction weighted**:

$$C_{\text{cell}} = f + (1 - f) C_s, \quad \text{rhs} = \frac{-\text{conv} + \text{diff} + f q_f + (1 - f) C_s q_s}{C_{\text{cell}}}, \quad (6)$$

with  $f$  the fluid volume fraction of the cell. The volumetric source is weighted with the same fraction, as the power density it is.

$f$  is pure geometry, built once at init beside  $\varphi$  from the closed plane-in-box form and stored as `vfrac`; it rides the snapshot so a diagnostic can state the invariant the solver actually conserves. The common approximation  $\text{clip}(\frac{1}{2} + \varphi_c/h)$  is not an approximation of that form but its degenerate limit — the one a grid-aligned wall lands in.

**Why the weighting is not cosmetic.** It cannot change any steady state, because it divides a residual that vanishes there. It is what makes the *transient* second order: with a pointwise capacity a cut cell carries one material’s entire heat capacity, an  $O(1)$  error on an  $O(h)$  band, i.e. first order. Measured on a two-material eigenmode, against the pre-C3 binary on the same inputs:

|                             | field-error order | decay-rate order  |
|-----------------------------|-------------------|-------------------|
| fluid-fraction weighted (6) | <b>1.99, 2.00</b> | <b>1.99, 2.00</b> |
| pointwise capacity          | 1.71, 1.09        | 1.28, 0.54        |

### 4.4 What is deliberately absent

There is **no penalization** in the conjugate mode. Its solid cells are unknowns, not boundary values, so the volume-penalization term that the `dirichlet` mode applies is simply not entered. Likewise  $\nu_t$  is admitted neither in the solid nor at a cut face.

## 5 Time step

Two findings, both measured, are built into the explicit bound.

**The limit is not the maximum material diffusivity.** A cut face’s  $k_{\text{face}}$  can approach  $\max(\kappa_L, \kappa_R)$  while feeding the cell on the *other* side, whose capacity is the other material’s. A fluid cell against a  $\kappa_s = 1000$  solid therefore sees  $1000\times$  the fluid rate *even when*  $\alpha_s = \alpha_f$ . The rate is consequently assembled from the *actual face coefficients* rather than from  $\max(\kappa/C)$  — which is also some  $500\times$  less conservative at  $w = \frac{1}{2}$ .

**A cut cell attains the Gershgorin bound.** The uniform interior never excites  $\rho \leq 2A_{ii}/C_i$ ; a cut cell does. Its rate is therefore raised: the implementation uses

$$\text{share} = \begin{cases} 2 & \text{at a cut cell,} \\ 6 & \text{otherwise,} \end{cases}$$

i.e.  $dt \times \frac{2}{3}$  *at cut cells only*. The history is worth keeping: the first version used 6 everywhere, C1 halved it to 3, and C2 then measured the default sitting at 96% of the bound and found an oblique high-contrast case that went to NaN there. At  $\text{share} = 2$  the margin is  $1.56\times$  and that case runs.

## 6 Order of accuracy

The scheme is **second order**, and the condition is a geometric one: the local temperature gradient must be interface-*normal* at the cut face. That is exact when the interface normal is a grid direction, and approached in a resolved thermal layer at any orientation. Where it fails — an obliquely cut face carrying a genuine tangential gradient — the cut-face flux error does not converge at all. Each statement below is measured.

| regime                                                                                   | what is measured                                                                  | result                                                   |
|------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|----------------------------------------------------------|
| Interface normal along a grid direction, <i>any</i> cut position within the cell; steady | 1D two-material slab, cut swept through a full cell, $\kappa_s$ over five decades | <b>exact</b> , $\leq 9.7 \times 10^{-15}$                |
| The same, <i>transient</i>                                                               | decaying two-material eigenmode; field error and decay rate                       | <b>1.99, 2.00</b>                                        |
| The same, but with a <i>pointwise</i> cut-cell capacity                                  | the same eigenmode, archived pre-C3 binary                                        | 1.71, 1.09 — <b>first order</b>                          |
| Oblique face, gradient purely interface-normal ( $r = 0$ )                               | cut-face normal flux against the analytic conjugate field                         | <b>exact</b> , $\sim 10^{-10}$ (geometry floor)          |
| Oblique face, tangential ratio $r > 0$                                                   | the same, at $h = 1/64, 1/128, 1/256$                                             | error $\propto r$ , <b>independent of <math>h</math></b> |

**The two ways to be second order.** The series resistance (5) is the *exact* solution of the one-dimensional conjugate problem, so wherever the problem is locally one-dimensional the interface contributes no error at all and the scheme is simply the underlying second-order discretisation. Two distinct situations qualify, and the first is not an approximation:

- the interface normal is a grid direction — every flat-wall case, including all of the channel validation in §11. The cut position is irrelevant: the slab gate sweeps it through a whole cell and the error stays at round-off;
- the local gradient is interface-normal,  $r = |\nabla_t T|/|\partial_n T| \rightarrow 0$ , which is the resolved-DNS premise the scheme was designed on.

**Where it is not second order, and it is worse than first.** At an obliquely cut face the baseline’s implied normal flux is  $q_n + h_d s_t / (a\mathcal{R})$ , and since  $\mathcal{R} \sim h_d/k$  the error is  $\sim k s_t/a$  — proportional to a *physical* gradient and carrying no power of  $h$ . Measured at  $\theta = 30^\circ$ ,  $\kappa_s = 10$ , the rms flux error is  $3.56 \times 10^{-2}$ ,  $3.77 \times 10^{-2}$ ,  $3.59 \times 10^{-2}$  at  $h = 1/64, 1/128, 1/256$ : an observed order of  $-0.01$  across a fourfold refinement, i.e. flat. It scales exactly linearly with  $r$  ( $3.56 \times 10^{-2}$ ,  $3.56 \times 10^{-1}$ ,  $3.56$  at  $r = 0.01, 0.1, 1$ ).

So at such a face the error **plateaus at**  $O(r)$  rather than reducing. This is a consistency error on a codimension-one set, so how much of it survives into the solution norm is not settled here — the flux is conservative, and no global refinement study has been run for an oblique interface with  $r > 0$ . What is settled is the flux itself, and that it does not converge.

**Why that was accepted.** Because  $r \sim 10^{-2}$  in a resolved thermal layer, the plateau sits near 1%, and because the measured crossover is  $r^* \approx 0.027$ : below it the corrected flux is *worse* than the baseline (§9). Restoring formal second order at oblique faces is therefore not a matter of switching the existing term on.

**The capacity is what secures the transient order.** Note the third row of the table: the spatial treatment above is unchanged there, and the order still falls to first. A cut cell carrying one material’s whole heat capacity is an  $O(1)$  error on an  $O(h)$  band, and the decay rate  $\mu$  is a global functional of the capacity distribution, so it is visible in every norm.

## 7 Configuration surface

| key                                          | meaning                                                                    |
|----------------------------------------------|----------------------------------------------------------------------------|
| [scalar.N] <code>ibm_wall = conjugate</code> | selects the mode (beside <code>dirichlet</code> , <code>adiabatic</code> ) |
| <code>solid_k</code>                         | $\kappa_s$                                                                 |
| <code>solid_rhocp</code>                     | $C_s$                                                                      |
| <code>solid_init</code>                      | initial solid value                                                        |
| <code>solid_source</code>                    | volumetric source inside the solid                                         |
| <code>contact_resistance</code>              | $R_c$ in (5)                                                               |
| <code>tangential_correction</code>           | the C2 term, <b>default off</b> (§9)                                       |
| [scalar] <code>indicator_interval</code>     | the free error indicator                                                   |
| [scalar] <code>heat_interval</code>          | the interface-heat diagnostic                                              |

`dirichlet` and `adiabatic` were left untouched: re-expressing them through the new arithmetic could not have been bit-exact, and bit-exactness of the existing modes was worth more than the shared code path.

**Five hard configuration errors.** Each of these silently produces a wrong answer if allowed, so each stops the run with a message naming the fix: `remove_solid = true` (the solid must exist); `refine_body` without `keep_buried` (same); `ibm_value` on a conjugate scalar (there is no imposed value); a `solid_*` key on a non-conjugate scalar; no immersed body at all; and wall functions. A sixth check is geometric rather than configurational: the 2:1 precondition —

a cut face may not sit on a coarse/fine block face, because the coefficient is a same-level arm — is *verified* at init, not assumed.

## 8 Diagnostics

**Interface heat (the Nusselt number).** The conjugate branch sums the scheme’s *own* face flux over the cut faces, with the penalization column explicitly zero — there is none in this mode, and `coef_p` is finite in a graded fluid cell, so writing the zero explicitly forestalls a double count. Gated three ways: against a closed form ( $4.6 \times 10^{-13}$ ), against an independent Python transcription ( $3.6 \times 10^{-16}$ ), and against a control-volume budget with the flow on that falls like  $dt^2$  and does not move with `niter`.

**The error indicator.** `indicator_interval` reduces  $e_{\text{face}} = h_d s_t / (T_R - T_L)$  over the cut faces (max, rms, count). This is the closed form of the relative error the scheme makes by dropping the tangential term, so it is a genuine *a-posteriori* measurement rather than an assumption, and it is free. It ships enabled.

## 9 Present but disabled: the tangential correction

The exact cut-face flux carries a second term,  $s_t (k_{\text{loc}} - k_{\text{face}})$ , which the scheme above drops. It *is* implemented, as its own flux divergence added after the baseline term — so it is inert by construction when off, not by a `+0.0` argument — and it is **default off**. Three independent measurements decided that:

1. a crossover ratio  $r^* \approx 0.027$ , which the DNS-like  $|\nabla_t T|/|\partial_n T| \sim 10^{-2}$  sits below;
2. an  $r = \infty$  boundary-value problem in which the corrected *face flux* is exact to  $10^{-11}$  while the corrected *field* is  $2.6\times$  *worse* — a finite-difference divergence wants the face *average*, not the midpoint value;
3. decisively, a cylinder at  $\kappa_s = 10^3$  where the correction is  $40\text{--}65\times$  worse than dropping the term, because  $|\nabla_t T| \propto 2/(1 + \kappa_s) \rightarrow 0$  in the isothermal limit while the discrete estimate of it does not.

The measured fix for (3) —  $k_{\text{area}}$ , the face area-weighted mean, which removes the paradox exactly on a plane — is recorded but deliberately not implemented: it cannot rescue the feature, because the binding constraint is the  $s_t$  estimate on curved interfaces. **Anyone reopening this starts from  $s_t$ , not from the multiplier.**

## 10 Not implemented

For the avoidance of doubt, the following appear in the design note and do *not* exist in the code: the corner/wedge (COCO) model for conducting sharp corners; the escalation back to the full exact flux as a supported mode; and any pluggable local-solution interface. The corner model rests on the same pointwise-flux premise that measurement (2) above falsified, so it should not be started without first settling the  $s_t$  estimate.

## 11 Validation status

**Manufactured solutions and identities.** A 1D two-material slab with the cut swept through a full cell and  $\kappa_s$  over five decades:  $\max |\theta - \theta_{\text{exact}}| \leq 9.7 \times 10^{-15}$  over 28 pairs, and

$w$  itself  $2.2 \times 10^{-14}$  against the analytic cut, rebuilt from the case file without the solver. Capacity-independent steady state; contact resistance exact;  $\sum C\theta \Delta V$  drift  $1.2 \times 10^{-16}$  relative with the flow on. 1 rank = 4 ranks and CPU = GPU at `max_abs 0` on both geometry paths. Every `ibm_wall`  $\neq$  `conjugate` run and every `[scalar] count = 0` run bit-exact against the pre-feature binaries.

**Turbulence.** A conjugate channel at  $Re_\tau = 149$ ,  $Pr = 0.71$ , matched to Flageul et al. (2015) in thermal problem, Reynolds number, grid resolution ( $\Delta y^+ = 0.49 \rightarrow 4.87$ ,  $\Delta x^+ = 8.4$ ,  $\Delta z^+ = 4.2$ ), solid thickness ( $d^+ = 149$ ) and outer-wall boundary condition (imposed flux), compared against *their published raw data* over a clean 29 684 wall-unit window. Agreement, as a percentage of each quantity’s own peak:

|                   |       |                    |       |
|-------------------|-------|--------------------|-------|
| $U^+$             | 0.4 % | $-\overline{u'v'}$ | 0.4 % |
| $\theta^+$        | 1.2 % | $\overline{u'T'}$  | 1.3 % |
| $\overline{u'^2}$ | 0.9 % | $\overline{v'^2}$  | 1.8 % |
| $\overline{w'^2}$ | 3.8 % | $\overline{T'^2}$  | 2.4 % |

The near-wall variance peak is 5.98 against their 5.94, +0.7 %, and  $\theta_\tau$  is  $\kappa_s$ -independent across the sweep to 0.003 %. The velocity comparison matters independently: it is the first against a trustworthy reference, so the thermal agreement is not resting on a compensating error in the flow.

**Inside the solid.** The scheme is also checked where the solid is the answer rather than a boundary condition — the variance profile *through* the slab, against their `fluctb` data:

| depth $y^+$ | −77    | −102   | −145 (outer face) |
|-------------|--------|--------|-------------------|
| ours        | 0.0113 | 0.0075 | 0.0063            |
| theirs      | 0.0113 | 0.0083 | 0.0073            |

This one carries a methodological warning worth more than the numbers. The first attempt read  $8\times$  their value at the outer face, and the cause was neither the scheme nor the grid: the solid diffusion time is  $d^2/\alpha_s \approx 106$  time units and the averaging window was shorter than *one*, so the statistics were dominated by the initial transient. Splitting the variance into a within-plane (spatial) and a plane-mean (temporal) part isolates it — the temporal term falls by a factor 45 when the transient is dropped. **Start conjugate statistics only after one solid diffusion time, and seed the solid rather than waiting for it:** with an imposed flux at both outer faces the solid’s mean level is a pure-Neumann null-space mode, so it can be set in closed form instead of relaxed.